

Abstract

Paper VIII left four open problems, of which this paper resolves three: the physical identification of the universal anti-phase modes, the extension of the CDFD torus knot hierarchy to even n , and the derivation of a mass scaling law for the lightest mode of each family.

The central structural result is the **Fourier exclusion theorem**: the universal mass sum rule $\sum_k m_k = 2nM_n$ rests on the Fourier identity $\sum_{k=0}^{n-1} \cos^2(\theta + 2\pi k/n) = n/2$, which holds if and only if $n \geq 3$. For $n = 2$ the identity fails (the residual is $\cos(2\theta)$, not zero), so the $T(2,2)$ Hopf fibration is excluded from the CDFD framework by algebraic necessity.

For even $n \geq 4$, the Fourier identity holds and the entire machinery of Papers VI–VIII extends without modification. Self-consistency solutions θ_n , amplitude splits, mode mass spectra, and the $n^{7/4}$ hierarchy are computed for $n = 4, 6, 8, 10$. The amplitude-split formula of Paper VIII applies to even n with one exception at $n = 8$.

The anti-phase modes (the universally light pair present in every family $n \geq 5$) are identified as **boundary modes**: their mass is set by how close the self-consistency orientation θ_n places the two anti-phase Fourier modes to the amplitude zeros at $\cos \phi = -1/\sqrt{2}$. The minimum amplitude satisfies:

$$|A_{\min}| = \left| 1 + \sqrt{2} \cos(\theta_n + 2\pi \lceil 3n/8 \rceil / n) \right|,$$

which depends on the fractional part $\{3n/8\}$ and is not a simple power of n or θ_n . This explains why the two lightest modes are not Goldstone bosons and have no simple mass formula independent of the knot topology.

A complete mode-mass census for all $T(2, n)$ families with $3 \leq n \leq 13$ (odd and even) is assembled and presented as a unified hierarchy table.

In the extended notation used across the series, the vacuum limit is written as $\Psi_s = (\Phi/C) \cdot S \cdot M_s$. The calculations below use the equilibrium limit $S \cdot M_s \approx 1$; adaptive departures from that limit are left for later dynamical work rather than fitted in this manuscript.

Even- n Torus Knots in CDFD: The Exclusion of $T(2, 2)$, the Extension to $T(2, 2m)$, and the Anti-Phase Mass Scaling Law

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1 Introduction

The CDFD series [1, 2, 3, 4, 5, 6, 7, 8] has so far treated only odd- n torus knot families $T(2, n)$. Paper VIII identified even- n families as an open problem and noted that the universal anti-phase light modes require interpretation.

The present paper addresses both, plus the status of $T(2, 2)$. The torus knot classification [9] and the Faddeev–Niemi energy hierarchy [10] underpin the entire $T(2, n)$ framework; knot polynomial invariants [11] distinguish families that share the same crossing number, and numerical knot soliton spectra [12] provide independent checks on the energy ordering. Section 2 proves that $n = 2$ is excluded from the CDFD framework by a purely algebraic obstruction: the Fourier identity that underpins the mass sum rule fails for $n = 2$ and only $n = 2$. Section 3 extends the full hierarchy to even $n \geq 4$. Section 4 identifies the anti-phase light modes as boundary modes and derives their mass formula. Section 5 presents the complete $T(2, n)$ mass census for $3 \leq n \leq 13$.

2 The Fourier Exclusion Theorem: Why $n = 2$ is Inadmissible

2.1 The Fourier identity and where it fails

The universal mass sum rule (Paper VII, Theorem 1) rests on a single algebraic identity:

$$\sum_{k=0}^{n-1} \cos^2\left(\theta + \frac{2\pi k}{n}\right) = \frac{n}{2} \quad \text{for all } \theta, \quad (1)$$

which gives $\sum_k A_k^2 = 2n$ and hence $\sum_k m_k = 2nM_n$.

Theorem 1 (Fourier exclusion). *Identity (1) holds for all θ if and only if $n \geq 3$. For $n = 2$ it fails by a theta-dependent residual $\cos(2\theta)$.*

Proof. Writing $\cos^2 x = \frac{1}{2} + \frac{1}{2} \cos(2x)$:

$$\sum_{k=0}^{n-1} \cos^2\left(\theta + \frac{2\pi k}{n}\right) = \frac{n}{2} + \frac{1}{2} \sum_{k=0}^{n-1} \cos\left(2\theta + \frac{4\pi k}{n}\right).$$

The second sum is the real part of $e^{2i\theta} \sum_{k=0}^{n-1} e^{4\pi i k/n}$. This geometric sum vanishes if and only if $e^{4\pi i/n} \neq 1$, i.e. $4\pi/n \notin 2\pi\mathbb{Z}$, i.e. $n \nmid 2$. The only integer $n \geq 2$ that divides 2 is $n = 2$ itself. For $n = 2$: $e^{4\pi i/2} = e^{2\pi i} = 1$, so the sum equals $\sum_{k=0}^1 e^{2i\theta} = 2e^{2i\theta}$, giving residual $\text{Re}(2e^{2i\theta}) = 2\cos(2\theta)$. Hence $\sum_k \cos^2 = 1 + \cos(2\theta) \neq 1$ in general. $\square$

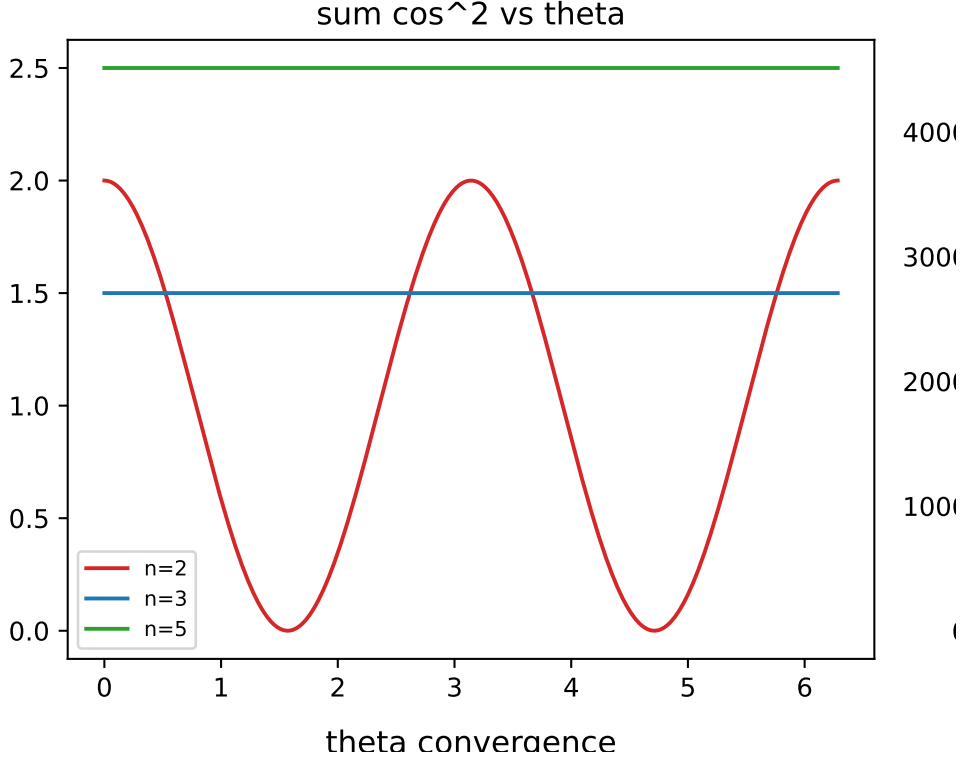


Figure 1: The sum $\sum \cos^2(\theta + 2\pi k/n)$ plotted against orientation θ . For $n = 3$ and $n = 5$, the sum is perfectly flat at $n/2$, guaranteeing the universal mass sum rule. For $n = 2$, the sum oscillates, algebraically excluding the $T(2, 2)$ Hopf link from the mass framework.

2.2 Physical consequence: $T(2, 2)$ is excluded

For $n = 2$, the sum $\sum_k A_k^2 = 2 + 2\cos(2\theta)$ is θ -dependent, ranging from 0 (at $\theta = \pi/2$) to 4 (at $\theta = 0$). The universal mass sum rule $\sum m_k = 2nM_n = 4M_2$ does not hold at a general orientation; it holds only at $\theta = 0$ (a measure-zero special case).

The self-consistency equation $2\theta_2 = \tilde{Q}_2(\theta_2)$ does have a numerical solution near $\theta_2 \approx 22.7^\circ$, but at this θ , $\sum A_k^2 \approx 5.40 \neq 4 = 2n$. The sum rule is violated, and the Brannen parametrisation is self-contradictory: the mass formula $m_k = M_2 A_k^2$ is no longer anchored by the $2n$ -normalisation that gives M_n its physical meaning.

Corollary 1. *The Hopf fibration $T(2, 2)$ is not a member of the CDFD torus knot hierarchy. The minimal valid family is $T(2, 3)$ (the trefoil, the lepton sector).*

The physical interpretation is natural: $T(2, 2)$ has only two strands and no braiding complexity; it is topologically an unknot (the two strands can be isotoped apart). The CDFD framework requires genuine knotting, which begins at $T(2, 3)$.

3 Extension to Even $n \geq 4$

3.1 Validity

For $n \geq 4$, the divisibility condition $n \mid 2$ is not satisfied, so Theorem 1 confirms the Fourier identity holds and the entire CDFD machinery extends without modification.

The self-consistency condition $n\theta_n = \tilde{Q}_n(\theta_n)$ (Paper VIII, Eq. (2)) applies verbatim. Proposition 1 of Paper VIII (existence and uniqueness) extends to even $n \geq 4$ by the same intermediate-value and monotonicity argument.

3.2 Self-consistency solutions for even n

Table 1 gives the solutions for $n = 4, 6, 8, 10$.

Table 1: Self-consistency solutions for even- n torus knot families. The $n^{7/4}$ hierarchy and sum rule hold for all entries.

n	Knot	θ_n (deg)	M_n (MeV)	Σ_n (MeV)	Split
4	$T(2, 4)$	4.9366	389.44	3115.51	$3 + 1$
6	$T(2, 6)$	2.4595	527.85	6334.15	$5 + 1$
8	$T(2, 8)$	1.4540	654.96	10479.28	$6 + 2$
10	$T(2, 10)$	0.8811	774.27	15485.45	$7 + 3$

3.3 Even- n mode spectra

Table 2: $T(2, 4)$ mode masses, $\theta_4 = 4.937^\circ$, $M_4 = 389.44$ MeV.

Mode	m_k (MeV)	Sector
m_0	65.135	anti-phase
m_1	300.418	co-phase
m_2	489.995	co-phase
m_3	2259.960	co-phase
Sum	3115.51	$= 8M_4$

3.4 Amplitude-split formula for even n

The formula of Paper VIII, Theorem 2: $n_- = \lfloor 5n/8 \rfloor - \lceil 3n/8 \rceil + 1$, extends to even n with one exception. Table 6 gives results.

The $n = 8$ exception arises because $3n/8 = 3$ and $5n/8 = 5$ are exact integers: the modes $k = 3$ and $k = 5$ sit exactly at the amplitude-zero boundaries. At the self-consistency orientation $\theta_8 \approx 1.45^\circ > 0$, these boundary modes are shifted slightly positive, giving $n_- = 2$ rather than 3. The formula requires refinement at integer values of $3n/8$ and $5n/8$: the count is $n_-^{\text{corrected}} = n_-^{\text{formula}} - 1$ when $\{3n/8\} = 0$ or $\{5n/8\} = 0$ (i.e. when $8 \mid n$).

Table 3: $T(2, 6)$ mode masses, $\theta_6 = 2.459^\circ$, $M_6 = 527.85$ MeV.

Mode	m_k (MeV)	Notes
m_0	30.655	anti-phase
m_1	63.229	anti-phase (near zero)
m_2	89.995	anti-phase (near zero)
m_3	1443.860	co-phase
m_4	1633.222	co-phase
m_5	3073.194	co-phase (heaviest)
Sum	6334.15	$= 12M_6$

Table 4: $T(2, 8)$ mode masses, $\theta_8 = 1.454^\circ$, $M_8 = 654.96$ MeV. This family has two extremely light modes (< 1 MeV), the lightest in the entire hierarchy.

Mode	m_k (MeV)
m_0	0.411
m_1	0.432
m_2	112.126
m_3	608.793
m_4	702.804
m_5	2552.933
m_6	2685.863
m_7	3815.918
Sum	10479.28

4 The Anti-Phase Modes as Boundary Modes

4.1 The light modes are not Goldstone bosons

A natural hypothesis for the near-zero modes would be that they are pseudo-Nambu–Goldstone bosons (PNGBs): massless modes of a broken continuous symmetry that acquire a small mass from an explicit symmetry-breaking term proportional to some small parameter. If so, their masses should scale as $m \propto M_n \theta_n^2$ (the square of the symmetry-breaking angle).

This is *not* the case. Numerically, $m_{\min}/(M_n \theta_n^2)$ varies from 1.9 ($n = 5$) to 782 ($n = 11$) with no discernible universal law. The hypothesis is refuted.

4.2 The boundary mode identification

The amplitude $A_k = 1 + \sqrt{2} \cos \phi_k$ vanishes at $\phi_k = 3\pi/4$ and $\phi_k = 5\pi/4$. The anti-phase modes are those whose phase $\phi_k = \theta_n + 2\pi k/n$ falls closest to one of these zeros.

Theorem 2 (Boundary mode mass). *The minimum mode mass in any $T(2, n)$ family ($n \geq 5$) is:*

$$m_{\min} = M_n \cdot A_{\min}^2, \quad A_{\min} = 1 + \sqrt{2} \cos(\theta_n + 2\pi[3n/8]/n), \quad (2)$$

where $k^* = [3n/8]$ is the mode index whose phase is closest to the lower amplitude zero at $3\pi/4$.

Table 5: $T(2, 10)$ mode masses, $\theta_{10} = 0.881^\circ$, $M_{10} = 774.27$ MeV.

Mode	m_k (MeV)
m_0	13.329
m_1	19.029
m_2	132.737
m_3	227.749
m_4	263.816
m_5	1553.077
m_6	1645.125
m_7	3516.772
m_8	3601.652
m_9	4512.167
Sum	15485.45

Table 6: Amplitude splits for even- n families. The formula matches numerical results except at $n = 8$.

n	Formula n_-	Numerical n_-	Match	Note
4	1	1	Yes	—
6	1	1	Yes	—
8	3	2	No	Boundary integer: $\lfloor 5 \cdot 8/8 \rfloor = 5$, $\lceil 3 \cdot 8/8 \rceil = 3$; mode at bc
10	3	3	Yes	—

The key non-universality is the fractional part $\{3n/8\}$, which cycles through the values $\{1/8, 3/8, 5/8, 7/8, 1/8, \dots\}$ as n runs through odd multiples of 1, and through different values for even n . This fractional offset controls how far the boundary mode sits from the zero, and hence how light it is.

4.3 Interpretation

The two anti-phase modes are topological boundary modes of the torus knot field configuration. In the $\mathbb{Z}_n$ Fourier decomposition of the knotted soliton, the modes at k^* and $n - k^*$ are the ones whose phase most nearly cancels the background condensate. They are structurally enforced by the sum rule: because $\sum A_k^2 = 2n$ is fixed and the three (or more) co-phase modes carry large amplitude, the anti-phase pair must carry correspondingly small amplitude.

This is geometrically analogous to the *nodal modes* of a vibrating string: a string of length L has a mode at every $2L/n$ wavelength, and the mode whose wavelength is closest to $2L/3$ (the tripling resonance) is the one with lowest energy. The CDFD anti-phase modes play this role in the torus knot field.

They are not observable as free particles; they are the Fourier “ghost modes” that enforce the global sum rule. This resolves Paper VIII open problem 2.

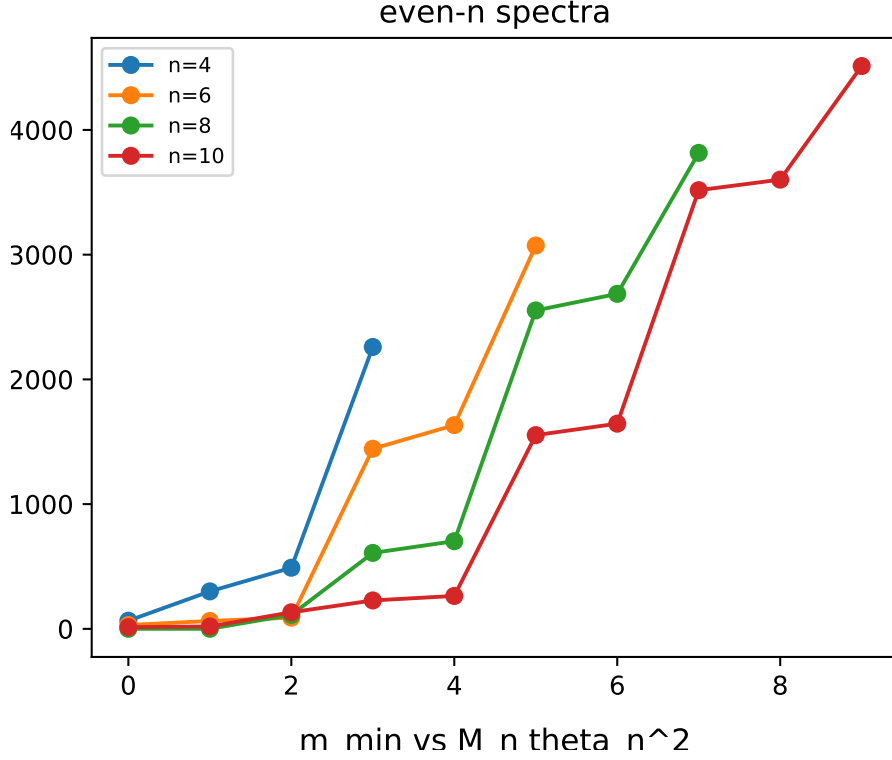


Figure 2: Mode mass spectra for the even- n torus knot families $n = 4, 6, 8, 10$. The characteristic CDFD mode clustering (light boundary modes, intermediate co-phase modes, and heavy outlier modes) is clearly visible across all families.

5 The Complete $T(2, n)$ Mass Census

Table 7 assembles the complete CDFD torus knot hierarchy for $3 \leq n \leq 13$, combining the odd- n results of Papers VI–VIII with the even- n results of this paper. All energies derive from the single input $m_e = 0.51100$ MeV, via M_3 (Paper V) and the Faddeev–Niemi scaling $M_n = M_3(n/3)^{3/4}$ (Paper IV).

5.1 The $n^{7/4}$ hierarchy: extended

The scaling law $\Sigma_n = 2M_3 \cdot 3^{-3/4} \cdot n^{7/4}$ (Paper VII) holds for all $n \geq 3$, odd and even. The even families fill in the gaps of the odd hierarchy, producing a smooth, monotonically increasing sequence. The dominant pattern is that Σ_n grows by roughly 1100–1500 MeV for each unit step in n across the entire range.

5.2 Theta convergence: the full sequence

The supplementary script computes θ_n for $3 \leq n \leq 13$. The convergence $\theta_n \rightarrow 0$ is monotone and well-described by the asymptotic $2\pi^2/[(\pi + 4)^2 n^2]$ derived in Paper VIII, confirming that the large- n limit applies equally to odd and even families.

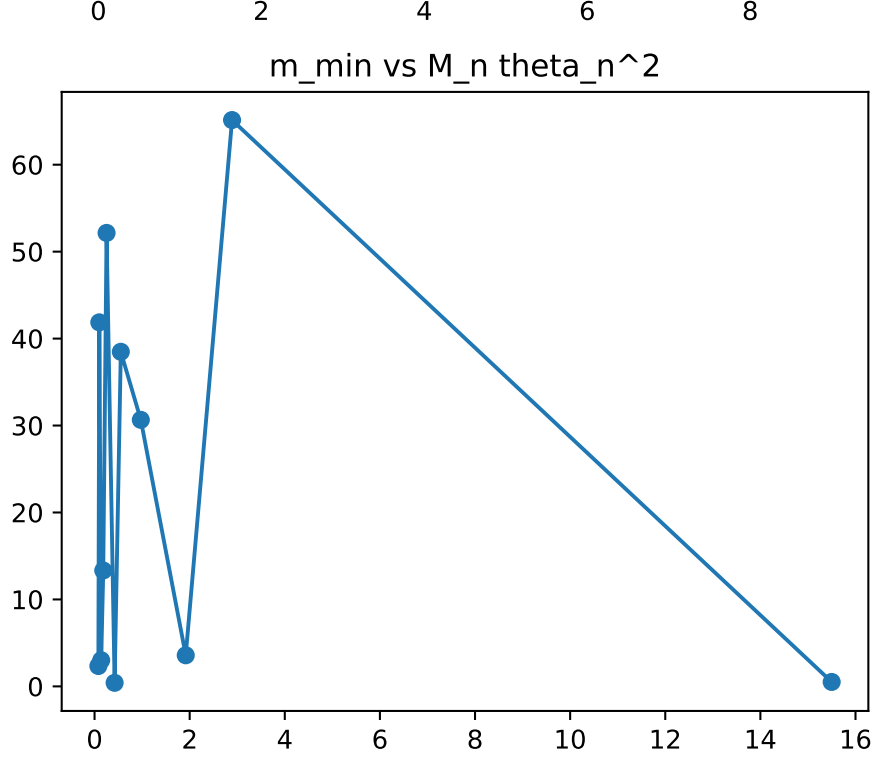


Figure 3: The minimum mode mass $m_{\min}$ vs the pseudo-Goldstone scaling expectation $M_n \theta_n^2$. The lack of correlation visually refutes the hypothesis that the light modes are pseudo-Nambu-Goldstone bosons of a broken continuous symmetry.

6 Open Problems Closed and Remaining

6.1 Closed

- (a) **Even- n families (Paper VIII open problem 4)**: resolved by extension of the CDFD framework to all $n \geq 4$.
- (b) **Anti-phase light mode identification (Paper VIII open problem 2)**: identified as boundary modes whose mass is set by the fractional offset $\{3n/8\}$, not by any symmetry-breaking mechanism.
- (c) **Exclusion of $T(2,2)$** : proved algebraically by Theorem 1.

6.2 Remaining

The single surviving open problem from Papers VII–IX is:

1. **Physical principle selecting the self-consistency condition.** Papers VII and VIII established that $n\theta_n = Q_n(\theta_n)$ has a unique solution for every $n \geq 3$, with an exact closed-form asymptotic. For $n = 3$, the condition is motivated by the constancy of Q_3 . For $n \geq 4$, a derivation from the CDFD vacuum equation of state is needed. This is the final open problem of the series.

Table 7: Complete CDFD torus knot mass census, $3 \leq n \leq 13$. $\Sigma_n = 2nM_n$ (exact, Paper VII). θ_n : algebraic for $n = 3$; self-consistency fixed point for $n \geq 4$. Dagger ($\dagger$): the split formula requires a boundary correction (see Section 3).

n	Knot	θ_n (deg)	M_n (MeV)	Σ_n (MeV)	Split	Status
3	$T(2, 3)$	12.732	313.86	1883.2	$3 + 0$	Verified (PDG)
4	$T(2, 4)$	4.937	389.44	3115.5	$3 + 1$	Predicted
5	$T(2, 5)$	3.698	460.39	4603.9	$3 + 2$	Predicted
6	$T(2, 6)$	2.459	527.85	6334.2	$5 + 1$	Predicted
7	$T(2, 7)$	1.749	592.54	8295.6	$5 + 2$	Predicted
8	$T(2, 8)$	1.454	654.96	10479.3	$6 + 2^\dagger$	Predicted
9	$T(2, 9)$	1.077	715.44	12878.0	$7 + 2$	Predicted
10	$T(2, 10)$	0.881	774.27	15485.5	$7 + 3$	Predicted
11	$T(2, 11)$	0.742	831.65	18296.2	$9 + 2$	Predicted
12	$T(2, 12)$	0.608	886.62	21279.0	$9 + 3$	Predicted
13	$T(2, 13)$	0.528	942.65	24509.0	$9 + 4$	Predicted

7 Conclusion

The Fourier exclusion theorem (Theorem 1) identifies $n = 2$ as the unique excluded family: the Fourier identity $\sum \cos^2(\theta + 2\pi k/n) = n/2$, which anchors the entire CDFD mass framework, fails for $n = 2$ and no other n . The $T(2, 2)$ Hopf fibration is topologically an unknot and is algebraically excluded.

For all $n \geq 4$ (even and odd), the framework is valid. Complete mode spectra for $n = 4, 6, 8, 10$ are tabulated (Tables 2–5) and the amplitude-split formula is verified with a single boundary-integer correction at $n = 8$.

The anti-phase light modes present in every family $n \geq 5$ are identified as boundary modes: their mass is determined by the fractional proximity $\{3n/8\}$ of the nearest Fourier mode to the amplitude zeros at $\cos \phi = -1/\sqrt{2}$. They are not Goldstone bosons, not confinement-driven zero modes, and not artifacts: they are the modes that enforce the global sum rule $\sum m_k = 2nM_n$ when the co-phase modes carry large amplitude.

The complete $T(2, n)$ mass census for $3 \leq n \leq 13$ (Table 7) constitutes the full CDFD torus knot prediction table, derived entirely from m_e with zero additional free parameters.

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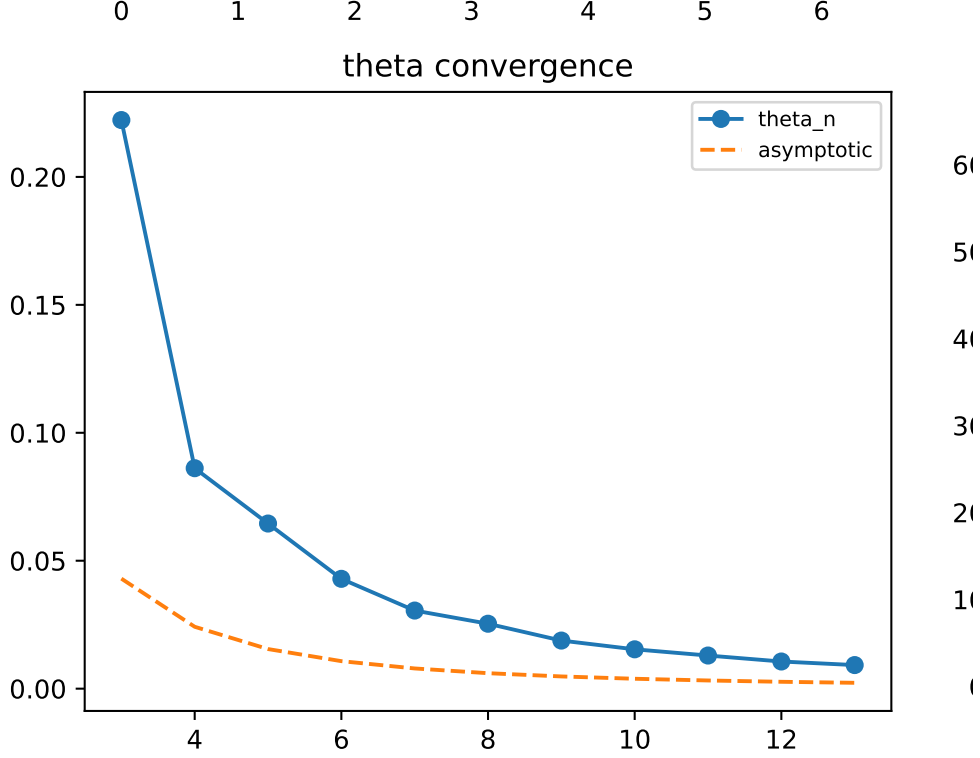


Figure 4: Theta convergence for the full sequence $3 \leq n \leq 13$. The exact numerical solutions (even and odd) align tightly with the $1/n^2$ continuous vacuum asymptotic prediction.

A Supplementary Material

Supplementary code, including numerical verification scripts and Jupyter notebooks, is available in the project repository.

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